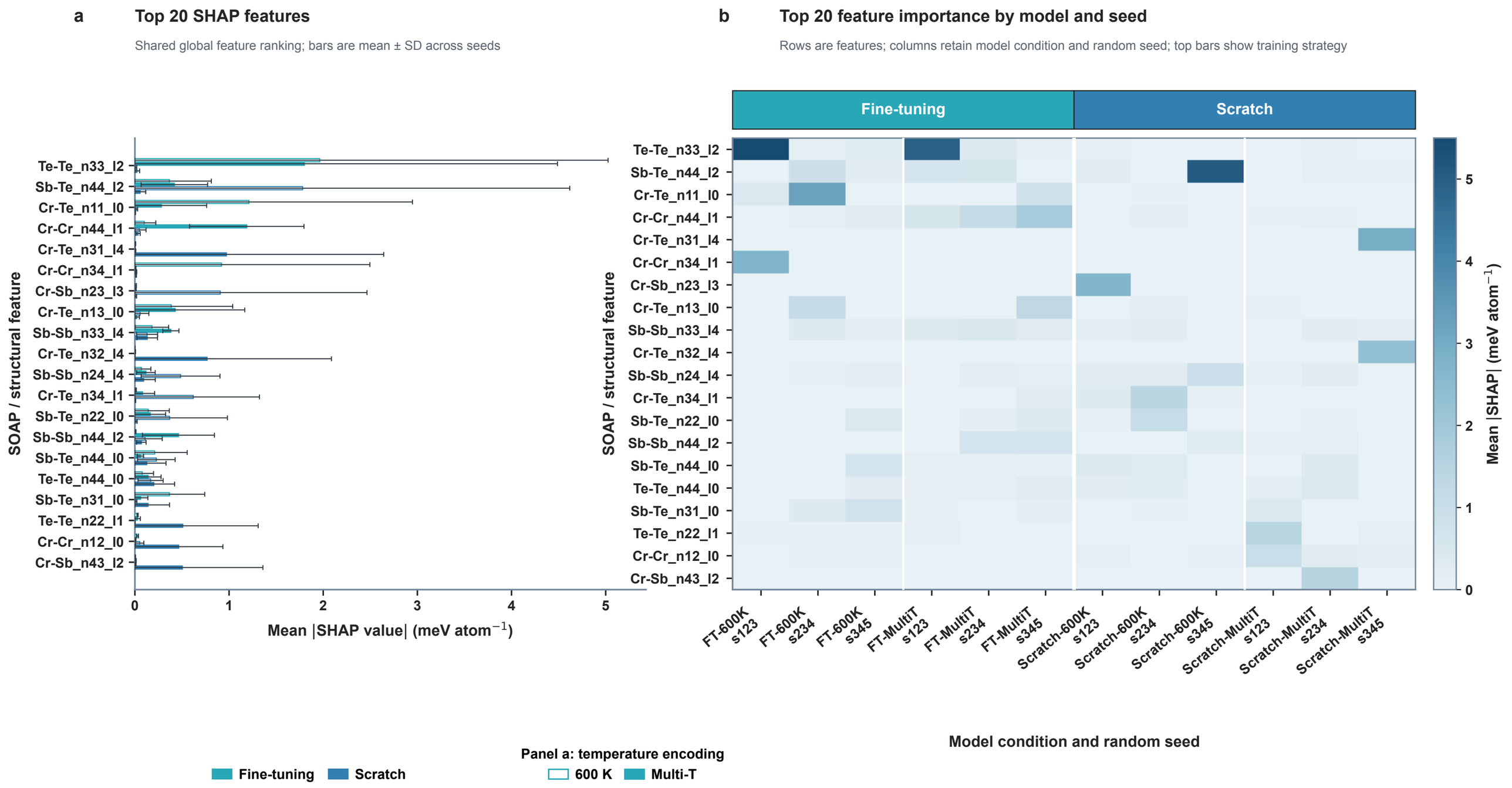

a
Top 20 SHAP features
Shared global feature ranking; bars are mean ± SD across seeds
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Te_n22_l0
Sb-Sb_n44_l2
Sb-Te_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
SOAP / structural feature
0
1
2
3
4
5
Mean |SHAP value| (meV atom^{-1})
b
Top 20 feature importance by model and seed
Rows are features; columns retain model condition and random seed; top bars show training strategy
Fine-tuning
Scratch
SOAP / structural feature
FT-600K s123
FT-600K s234
FT-600K s345
FT-MultiT s123
FT-MultiT s234
FT-MultiT s345
Scratch-600K s123
Scratch-600K s234
Scratch-600K s345
Scratch-MultiT s123
Scratch-MultiT s234
Scratch-MultiT s345
Model condition and random seed
Mean |SHAP| (meV atom^{-1})
Panel a: temperature encoding
Fine-tuning
Scratch
600 K
Multi-T